23/7/26
EX NO: 2

Types of network cables

Aim:

Study of different types of network cables.

a) understand different types of network cable.

different type of cables used in networking are:

1. unshilded twisted pair (UTP) cable
2. shilded twisted pair (STP) cable
3. Coaxial cable
4. Fibre optic cable.

Cable type	category	Maximum Data Transmission	Advantages/ Disadvantages	Application/ use
UTP	category 3	10 bps	advantages • cheaper in cost • Easy to install as they have a smaller overall diameter Disadvantages • More prone to (EMI) electromagnetic interference and noise	10Base-T Ethernet
	category 5	up to 100 Mbps		Fast Ethernet, Gigabit Ethernet
	category 5e	1Gbps		Fast Ethernet Gigabit Ethernet.
STP	category 6/6a	10 Gbps	Advantages • shielded • Faster than UTP. • Less susceptible to noise and interfac	Gigabit Ethernet (>1m) widely used in data centers

STP	category 7	10 Gbps	Disadvantages • Expensive • Greater installation effort	Gigabit Ethernet, 10G ethernet (55m) widely used in data centers
coaxial cable	RG-6 RG-59 RG-11	10-100 Mbps	• High bandwidth • Immune to interfac • Low loss bandwidth • versatile disadvantages • limited distance • cost • size is bulky	speed of signal is 500m Television network high speed internet connection
Fiber optics	single mode multi mode	100Gbps	Advantages • High speed • High bandwidth • High security • long distance disadvantages • Expensive • Required skilled installers	Maximum distance of fiber optics cable is around 100 meters